WasteWise Scholar

AI-Powered Sustainable Study Assistant

IBM SkillsBuild | AICTE | 1M1B AI for Sustainability Internship

Project Title	WasteWise Scholar – AI-Powered Sustainable Study Assistant
Primary SDG	SDG 12 – Responsible Consumption and Production
Secondary SDGs	SDG 4 – Quality Education | SDG 13 – Climate Action
Program	IBM SkillsBuild | AICTE | 1M1B AI for Sustainability Internship
Approach	Low-code / No-code AI Prototype
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Internship Project Report – First Year B.Tech (AI & DS)

1. Project Title

WasteWise Scholar – AI-Powered Sustainable Study Assistant (Prototype)

Reducing Paper Waste in Education Through Intelligent Summarisation, Topic Extraction, and Print Recommendations

2. Abstract

WasteWise Scholar is an AI-powered web application prototype designed to help students make informed, environmentally conscious decisions about printing study materials. The tool allows students to upload a document or paste study text, which is then analysed to generate a concise summary, extract key topics, and produce a print recommendation — advising whether a document is best printed or reviewed digitally.

This project directly supports SDG 12 (Responsible Consumption and Production) by targeting one of the most overlooked sources of waste in educational institutions: routine and often unnecessary paper printing. It also aligns with SDG 4 (Quality Education) through AI-enhanced study support, and SDG 13 (Climate Action) by raising student awareness of the environmental cost of printing habits.

WasteWise Scholar is an internship prototype developed as part of the IBM SkillsBuild, AICTE, and 1M1B AI for Sustainability Internship by a first-year B.Tech student in Artificial Intelligence and Data Science. The current version is a functional front-end prototype demonstrating the complete user workflow. The project is designed for future integration with IBM watsonx.ai and IBM Granite models to enable real-time summarisation and topic extraction.

3. SDG Alignment

WasteWise Scholar is purposefully designed around three United Nations Sustainable Development Goals:

SDG	Goal	Relevance to WasteWise Scholar
SDG 12	Responsible Consumption & Production	Targets reduction of unnecessary paper printing in academic settings
SDG 4	Quality Education	AI-powered tools support student learning and comprehension
SDG 13	Climate Action	Reducing print waste can lower the carbon footprint of education

> *Paper consumption in educational institutions is a significant and often invisible contributor to resource waste. WasteWise Scholar is designed to make this impact visible and personal — helping each student understand how their daily printing choices contribute to a larger environmental picture.*

4. Problem Description

Educational institutions consume large volumes of paper every academic year. Students routinely print full textbook chapters, lecture slides, lab manuals, and research papers — often reading only a portion of the printed content. This behaviour is rarely driven by necessity; it is mostly driven by habit and the absence of a convenient alternative.

Key factors contributing to this problem include:

- No quick way to preview a document's key content before making a printing decision
- Lack of personalised guidance on whether a document's content density justifies printing
- Limited awareness among students of the cumulative environmental cost of routine printing
- Over-reliance on printed notes even when digital alternatives are readily accessible
- No single tool that combines summarisation, topic extraction, and a print recommendation for student use

While it is difficult to quantify the exact scale of this problem without a dedicated study, it is widely acknowledged in environmental literature that paper consumption in education represents a meaningful and avoidable source of waste. WasteWise Scholar is motivated by this observation and aims to provide a practical, student-friendly tool to address it.

5. Problem Statement

> ***"Students lack an intelligent, accessible tool that helps them make informed, sustainability-conscious decisions about printing study materials — resulting in avoidable paper waste, unnecessary resource consumption, and missed opportunities to develop greener study habits."***

6. Target Users

WasteWise Scholar is designed primarily for students who regularly download and print academic material. The intended users include:

- Undergraduate students across all disciplines who print lecture slides, notes, and textbooks
- School students in Grades 9–12 preparing for board examinations with large volumes of material
- Self-learners and online course participants managing downloaded reading material
- Teaching faculty interested in promoting digital-first study habits in their classrooms
- Educational institutions aiming to reduce their paper footprint as part of sustainability goals

The primary user persona is a first- or second-year engineering student who downloads lecture PDFs and lab manuals and defaults to printing them without first assessing content density. WasteWise Scholar is built to fit naturally into this workflow.

7. Existing Challenges

Several gaps in existing tools and practices make this problem difficult to address without a dedicated solution:

- No existing student-facing tool combines summarisation, topic extraction, and a print recommendation in a single interface
- General-purpose AI summarisers are not tailored to academic document types such as lab records, textbooks, or question papers
- Sustainability metrics related to printing are rarely made visible or personal for individual students
- Students without programming experience cannot easily access NLP APIs or text-processing libraries
- Print decisions are often habitual and unconscious — requiring a direct, contextual nudge rather than general awareness campaigns
- Many low-code platforms available to students do not integrate easily with document-based workflows

8. Design Thinking Process

WasteWise Scholar was developed by working through the five stages of the Design Thinking methodology. As a first-year internship project, the process was carried out at a student scale — involving personal observation, informal peer conversations, individual ideation, prototype development, and self-reflection.

Empathise	I observed and had informal conversations with classmates and seniors about their study and printing habits. Many students mentioned that they print full chapters or lecture slides without reading them first, often because they feel more comfortable studying from paper. This helped me understand that the problem is not just about awareness — it is about the absence of a tool that makes digital reviewing feel as convenient as printing. These observations were informal and intended to understand common study and printing habits rather than conduct a formal research study.
Define	Based on these observations, I defined the core user need as: Students need a quick, accessible way to understand what is truly essential in a document before deciding whether to print it. The problem was not a lack of willingness to save paper, but a lack of a practical decision-support tool at the right moment in the workflow. My 'How Might We' question became: How might we help students study smarter digitally, so that printing becomes a deliberate choice rather than a default habit?

Ideate	I brainstormed several possible solutions, including a print quota tracker, a keyword flashcard generator, and an eco-study dashboard. I eventually chose to combine the most useful elements into a single tool: an AI-powered assistant that summarises content, extracts key topics, and recommends whether a document is worth printing. I identified IBM watsonx.ai and IBM Granite models as the proposed AI backbone for this tool, in line with the IBM SkillsBuild program's focus on enterprise AI.
Prototype	I built a front-end prototype as a single-page web application using HTML, CSS, and JavaScript. The prototype demonstrates the complete user workflow: uploading a study material, viewing an AI-generated summary, seeing extracted topics, receiving a print recommendation, and checking the sustainability dashboard. In the current prototype, the AI features are simulated to demonstrate the concept. The proposed implementation would use IBM watsonx.ai with IBM Granite models via API calls to perform actual text summarisation and topic extraction.
Reflect & Iterate	After reviewing the prototype myself and sharing it with a few classmates informally, I received feedback that the print recommendation section needed a brief explanation of why a particular decision was made. I added a small 'Why this recommendation?' note below the score. I also improved the layout to be more mobile-friendly. No formal user testing or statistical usability evaluation was conducted at this stage, as this is an early-stage internship prototype.

9. Solution Overview

WasteWise Scholar is a browser-based web prototype that simulates the workflow of an AI-powered study assistant. It demonstrates how a student can upload or paste study material and receive a summary, topic list, print recommendation, and sustainability estimate — all in one interface.

The prototype demonstrates the following four-step workflow:

- Step 1 – Upload: The student uploads a study document (PDF or text) or pastes content directly into the interface
- Step 2 – Analyse: In the proposed implementation, IBM watsonx.ai with IBM Granite models would process the text to extract topics and generate a summary; the prototype demonstrates this output with representative results
- Step 3 – Recommend: A rule-based scoring function evaluates content density and topic richness to generate a print recommendation — Print It, Skim Digitally, or Skip Printing
- Step 4 – Dashboard: The student views their summary, key topics, recommendation, and a simple sustainability counter estimating pages and resources saved

The prototype requires no installation, no account creation, and no prior technical knowledge, making it accessible to students across all backgrounds and streams.

10. Key Features (Prototype)

The following features are visible and functional in the current prototype:

- Upload Study Material: Students can upload a PDF or paste text directly into the interface for analysis
- Smart Summary (Prototype Demonstration) : The prototype displays a structured summary of the uploaded content, demonstrating the intended output of IBM Granite-powered summarisation
- Topic Extraction (Simulated): Key topics and concepts from the document are extracted and displayed as tags, illustrating the proposed IBM watsonx.ai NLP output
- Print Recommendation: A rule-based function classifies the document as 'Print It', 'Skim Digitally', or 'Skip Printing' based on word count and content structure, with a brief explanation of the recommendation
- Sustainability Dashboard: A session-level counter displays estimated pages saved and a rough environmental figure (based on published average estimates per printed page), clearly presented as indicative rather than precise. For awareness and educational purposes only.

The following features are identified as proposed future enhancements, not present in the current prototype:

- Live AI integration via IBM watsonx.ai API for real-time summarisation and topic extraction

- Multilingual support for Tamil, Hindi, and other Indian languages
- LMS plug-in versions for Moodle, Google Classroom, or Microsoft Teams
- Institutional dashboard for aggregate printing reduction reporting
- Gamification elements such as eco-badges and leaderboards
- Offline NLP processing using TensorFlow.js for low-connectivity environments

11. AI Components – Proposed Implementation

The current prototype demonstrates the user interface and workflow concept. The following AI components represent the proposed technical implementation for a fully functional version of WasteWise Scholar:

- IBM watsonx.ai and IBM Granite Models (Proposed): Intended for document summarisation, topic extraction, and generating concise study insights from uploaded academic content.
- Print Worthiness Scoring (Rule-Based): Currently implemented in the prototype as a rule-based function using word count, paragraph count, and estimated content density. A future version could incorporate a lightweight ML classifier trained on academic document features.
- Sustainability Estimation: Environmental impact figures (estimated pages saved, CO_2 offset) in the prototype are based on published average benchmarks and are displayed as indicative estimates. These are not measured or validated values.

Features such as OCR (Optical Character Recognition), Watson Language Classifier, IBM Cloud Functions, and REST API backend deployment are proposed future components and are not present in the current prototype.

12. System Architecture (Proposed)

The architecture below describes the intended design for a fully implemented version of WasteWise Scholar. The current prototype implements the Input Layer and Output Layer; the Processing and Analytics layers represent the proposed implementation using IBM watsonx.ai.

Layer	Description
Input Layer	Student uploads a PDF or pastes raw text into the web interface
Processing Layer (Proposed)	IBM watsonx.ai with IBM Granite models would extract key topics and generate summaries
Decision Engine	A rule-based scoring function calculates content density and recommends print action
Output Layer	Summary, keyword tags, and print recommendation are displayed to the student

Sustainability Module	Session-level counters track estimated pages saved and display eco-impact metrics
UI / Dashboard	Responsive web interface (HTML + CSS + JavaScript) — no login required

In the proposed full implementation, data would flow from student input through IBM watsonx.ai for NLP processing, then through the scoring engine, and finally to the output interface. All components would communicate statelessly to preserve privacy. The current prototype simulates this flow without live API calls.

13. Responsible AI Considerations

13.1 Transparency

Every print recommendation in the prototype is accompanied by a brief explanation of the factors considered — such as document length and content density. Students are not presented with an unexplained AI decision.

13.2 Privacy

The prototype does not store, log, or transmit any student-uploaded content. All processing in the current version is handled locally in the browser. The proposed API-based implementation would follow a stateless, session-only processing model with no persistent storage of user documents.

13.3 Fairness and Accessibility

The tool is designed to work across common academic document types regardless of subject or author. The interface is built to be simple and accessible, with no technical knowledge required. Future versions aim to support multiple Indian languages to improve inclusivity.

13.4 Avoiding Over-Reliance

WasteWise Scholar provides recommendations, not instructions. Students retain full autonomy over their printing decisions. The interface includes a clear note that AI-generated summaries should supplement, not replace, a thorough reading of high-stakes academic content.

13.5 Honesty in Environmental Claims

Environmental impact figures displayed in the sustainability dashboard are based on broadly cited estimates (for example, the approximate CO2 and water associated with producing a single printed page). These are clearly labelled as estimated figures and are not presented as precise measurements. The prototype does not make validated claims about real-world environmental impact.

14. Expected Impact

If deployed and adopted consistently, WasteWise Scholar has the potential to:

- Encourage students to review content digitally before deciding to print, reducing unnecessary paper use
- Raise individual student awareness of the environmental cost of routine printing through a personalised sustainability counter
- Support a gradual shift from print-first to digital-first study habits through consistent, contextual AI nudging
- Provide educational institutions with a lightweight tool to incorporate sustainability into everyday student workflows
- Demonstrate to students how AI can be applied to real-world SDG challenges as part of their own learning journey

Specific quantitative impact estimates (such as percentage reductions in paper use or CO2 savings per student) are not claimed at this prototype stage, as no deployment or longitudinal study has been conducted. These would need to be measured through actual use over time.

15. Future Scope

The following enhancements are planned for future development iterations of WasteWise Scholar:

- Live IBM watsonx.ai Integration: Replace the current simulated outputs with real-time API calls to IBM watsonx.ai and IBM Granite models for actual summarisation and topic extraction
- Multilingual Support: Extend the tool to support Tamil, Hindi, and other regional Indian languages to address SDG 4 for students studying in non-English mediums
- LMS Integration: Develop plug-in versions for Moodle, Google Classroom, and Microsoft Teams to embed the tool directly into student workflows
- Institutional Dashboard: Build an admin-facing view for colleges to track aggregate session data on documents analysed and estimated print reduction
- Gamification: Add eco-badges and milestone trackers to encourage consistent engagement with the tool
- Offline NLP: Explore TensorFlow.js or similar on-device models to provide basic functionality in low-connectivity environments
- Progressive Web App: Package the tool as a PWA for seamless use on mobile devices without installation
- User Testing: Conduct structured usability testing with a larger student group to validate the design and measure real-world impact

16. Conclusion

WasteWise Scholar is an internship prototype that demonstrates how AI can be applied to a practical, everyday sustainability challenge — the habit of unnecessary printing in educational institutions. As a first-year B.Tech student in Artificial Intelligence and Data Science, this project gave me the opportunity to work through the full Design Thinking process, explore the capabilities of IBM watsonx.ai and IBM Granite models, and build a functional web prototype that communicates a real SDG-aligned idea.

The prototype is honest in its current state: it demonstrates the intended workflow and user experience without making claims of production deployment, live AI integration, or validated impact metrics. The proposed use of IBM watsonx.ai and IBM Granite models represents a realistic and technically grounded path forward for a fully functional implementation.

I believe that tools like WasteWise Scholar — even at the student prototype stage — show how AI can be directed toward meaningful sustainability goals. SDG 12, SDG 4, and SDG 13 are not abstract targets; they are outcomes that everyday choices, including how students decide to study, can contribute to.

—— *End of Report* ——

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